

TROIS-RIVIERES, QC, Canada

WMO: 717240

Lat: 46.3537N

Lon: 72.5155W

Elev: 20

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-6.1	-20.6	1.8	-9.2	-15.5	2.4	-4.9	29.1	27.8	25.4	24.8	6.7	290	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	81.3	70.3	79.2	69.4	77.2	68.3	73.2	78.4	71.7	76.8	70.3	75.3	10.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.4	115.8	76.6	69.9	110.0	75.3	68.3	104.2	74.0	36.9	78.7	35.5	77.0	34.3	75.3	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	20.8	18.5	-16.5	86.3	5.6	3.5	-20.5	88.9	-23.7	90.9	-26.9	92.9	-30.9	95.4
			-16.6	75.7	5.3	2.0	-20.5	77.1	-23.6	78.3	-26.6	79.4	-30.5	80.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	14.1	17.0	26.8	40.2	54.5	64.4	69.4	67.9	60.2	47.9	35.1	22.7
	DBStd	20.95	12.15	10.64	9.64	7.02	6.70	5.74	4.41	4.69	6.50	6.97	8.31	10.41
	HDD50	4512	1114	924	719	302	30	0	0	0	3	124	449	847
	HDD65	8208	1579	1344	1184	745	331	78	10	26	172	530	897	1312
	CDD50	2138	0	0	0	7	170	433	603	554	309	60	2	0
	CDD65	358	0	0	0	0	6	62	148	114	27	1	0	0
	CDH74	1347	0	0	0	0	24	218	599	426	80	0	0	0
	CDH80	131	0	0	0	0	3	23	69	31	5	0	0	0
Wind	WSAvg	8.4	8.8	8.9	8.8	8.8	8.7	7.9	7.9	7.6	7.8	8.5	8.6	8.5
Precipitation	PrecAvg	39.9	2.8	2.2	2.7	3.1	3.3	3.8	4.2	3.9	3.7	3.7	3.2	3.4
	PrecMax	46.8	5.2	4.6	4.8	6.5	5.4	8.1	7.8	7.7	7.2	7.2	5.5	6.9
	PrecMin	32.1	1.3	1.1	0.9	1.1	1.4	1.7	1.7	1.5	2.0	1.1	0.8	1.1
	PrecStd	4.2	1.1	0.8	1.0	1.3	1.1	1.4	1.5	1.2	1.3	1.4	1.0	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	40.5	49.0	63.8	77.0	82.8	85.0	83.2	79.9	69.9	58.0	44.2
		MCWB	38.9	37.4	41.6	51.6	62.2	70.0	70.7	71.3	70.3	61.6	53.9	42.0
	2%	DB	36.9	37.4	43.5	57.8	72.1	79.1	81.7	80.5	76.4	65.0	52.5	40.5
		MCWB	35.5	35.0	38.3	48.4	59.4	67.2	70.9	70.1	68.7	59.4	49.3	38.4
	5%	DB	34.4	35.2	40.6	53.4	68.6	76.4	79.5	78.4	73.2	61.3	48.9	37.9
		MCWB	33.2	33.4	36.4	45.6	58.5	66.0	70.1	69.5	66.4	56.1	45.5	36.2
	10%	DB	31.6	32.9	38.1	50.0	65.1	73.6	77.3	76.3	70.5	58.0	45.9	35.7
		MCWB	30.4	31.3	34.7	43.4	56.1	64.7	69.0	68.3	64.4	53.5	43.0	34.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	38.4	43.8	54.9	67.0	72.7	75.6	75.0	72.4	64.2	54.9	42.7
		MCDB	39.9	39.4	45.8	60.2	73.8	79.2	81.5	80.6	77.6	67.9	56.5	43.7
	2%	WB	35.8	35.3	39.3	49.8	63.1	70.5	73.7	72.9	70.2	60.0	50.0	39.0
		MCDB	36.7	36.7	42.6	54.9	68.3	76.3	78.8	78.0	74.6	63.9	52.1	40.1
	5%	WB	33.4	33.6	37.0	46.9	60.4	68.7	72.1	71.2	67.9	56.9	46.1	36.5
		MCDB	34.3	35.0	39.6	52.2	65.9	73.9	77.0	76.1	72.4	60.4	48.2	37.7
	10%	WB	30.4	31.4	35.3	44.3	57.8	66.9	70.7	69.8	65.4	54.3	43.4	34.2
		MCDB	31.7	32.9	37.5	48.8	63.6	71.7	75.7	74.7	69.6	57.5	45.7	35.5
Mean Daily Temperature Range	5% DB	MDBR	13.9	15.0	13.6	13.6	15.3	14.3	13.5	13.9	14.1	11.6	9.7	10.7
		MCDBR	14.6	12.9	13.8	19.9	20.2	17.2	15.7	15.4	16.0	16.1	13.5	10.4
		MCWBR	14.6	12.0	10.0	12.9	10.7	9.3	8.9	9.2	10.4	11.5	12.1	10.1
	5% WB	MCDBR	14.1	12.5	12.7	18.2	17.0	14.6	13.7	13.7	14.5	14.6	13.1	10.1
		MCWBR	14.2	12.1	9.9	13.0	11.1	9.3	8.8	9.3	10.3	11.3	12.3	10.2
Clear-Sky Solar Irradiance	taub	0.262	0.285	0.318	0.350	0.374	0.414	0.417	0.407	0.374	0.336	0.308	0.279	
	taud	2.328	2.246	2.280	2.378	2.378	2.263	2.267	2.301	2.381	2.492	2.485	2.347	
	Ebn at Noon	268	282	288	288	283	271	268	266	266	262	249	246	
	Edh at Noon	23	31	36	36	37	42	41	38	32	25	21	21	
All-Sky Solar Radiation	RadAvg	450	734	1145	1423	1628	1781	1760	1543	1208	706	421	306	
	RadStd	62	82	103	143	170	130	156	72	98	56	54	25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page